



Universidad autónoma del Estado de
México

Ingeniería en computación

7mo semestre, 2025B

Sistemas operativos

Tarea: Llaves

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17/11/2025



1. Para comenzar crearemos las llaves usando el comando:

`gpg --full-gen-key`

No pedirá el tipo de llave que vamos a crear, en nuestro caso serán las opciones:

- 2 DSA and Elgamal
- 3 DSA (sign only)
- 4 RSA (sign only)
- 9 ECC (sign and encrypt)
- 10 ECC (sign only)

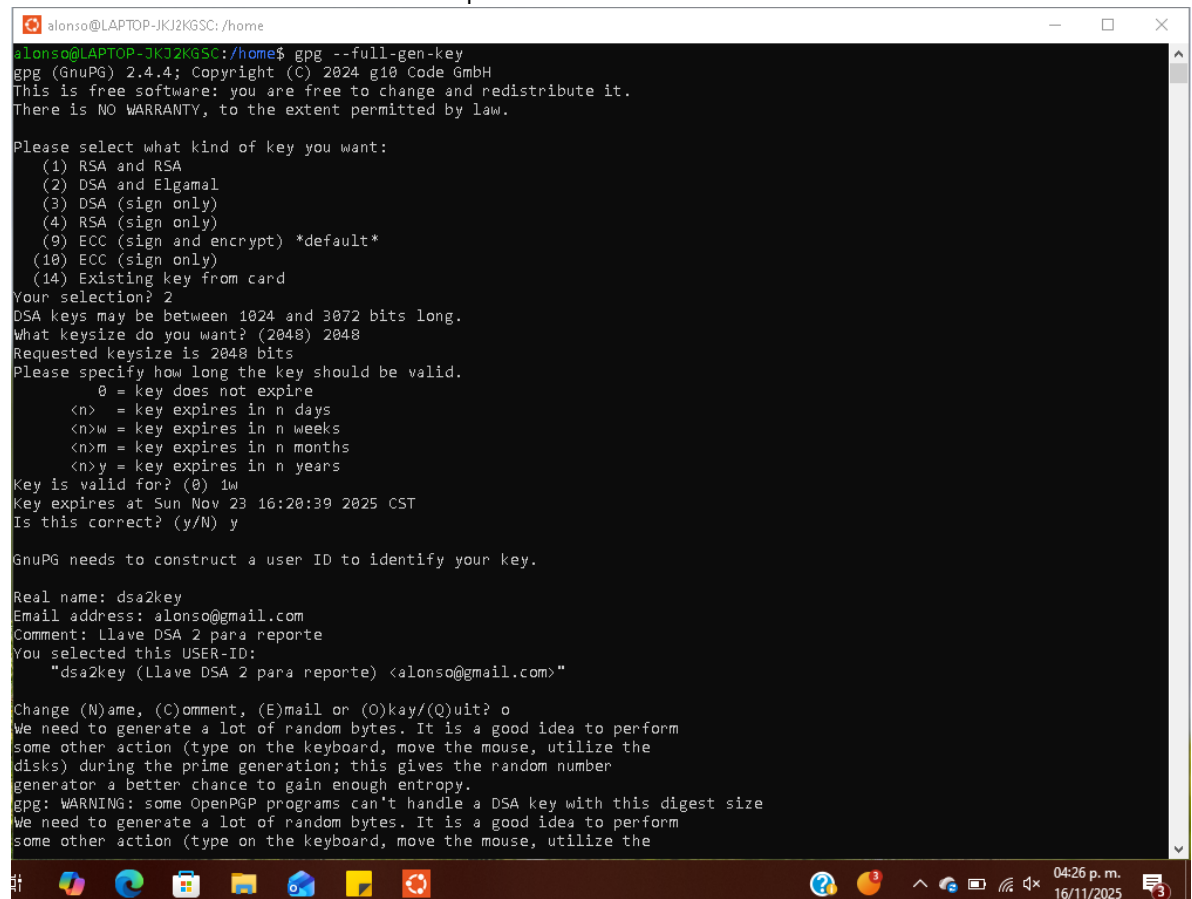
Después nos pedirá el tamaño en cuanto a bits, aquí va a variar según el tipo de llave que vayamos a crear.

También nos pedirá el tiempo de expiración de nuestras llaves, en nuestro caso pondremos 1w, que se refiere a una semana.

Confirmamos que todo es correcto.

Por último, nos pedirá nuestro nombre, correo, un comentario para identificar la llave.

Al final nos mostrara la llave que se acaba de crear.



```
alonso@LAPTOP-JKJ2KGSC:/home$ gpg --full-gen-key
gpg (GnuPG) 2.4.4; Copyright (C) 2024 g10 Code GmbH
This is free software: you are free to change and redistribute it.
There is NO WARRANTY, to the extent permitted by law.

Please select what kind of key you want:
  (1) RSA and RSA
  (2) DSA and Elgamal
  (3) DSA (sign only)
  (4) RSA (sign only)
  (9) ECC (sign and encrypt) *default*
  (10) ECC (sign only)
  (14) Existing key from card
Your selection? 2
DSA keys may be between 1024 and 3072 bits long.
What keysize do you want? (2048) 2048
Requested keysize is 2048 bits
Please specify how long the key should be valid.
  0 = key does not expire
  <n> = key expires in n days
  <n>w = key expires in n weeks
  <n>m = key expires in n months
  <n>y = key expires in n years
Key is valid for? (0) 1w
Key expires at Sun Nov 23 16:20:39 2025 CST
Is this correct? (y/N) y

GnuPG needs to construct a user ID to identify your key.

Real name: dsa2key
Email address: alonso@gmail.com
Comment: Llave DSA 2 para reporte
You selected this USER-ID:
  "dsa2key (Llave DSA 2 para reporte) <alonso@gmail.com>"

Change (N)ame, (C)omment, (E)mail or (O)kay/(Q)uit? o
We need to generate a lot of random bytes. It is a good idea to perform
some other action (type on the keyboard, move the mouse, utilize the
disks) during the prime generation; this gives the random number
generator a better chance to gain enough entropy.
gpg: WARNING: some OpenPGP programs can't handle a DSA key with this digest size
We need to generate a lot of random bytes. It is a good idea to perform
some other action (type on the keyboard, move the mouse, utilize the
```

public and secret key created and signed.

```
pub   dsa2048 2025-11-16 [SC] [expires: 2025-11-23]
      6BDB477415A49A3DF1BF7EF424E310D0490402C9
uid    dsa2key (Llave DSA 2 para reporte) <alonso@gmail.com>
sub    elg2048 2025-11-16 [E] [expires: 2025-11-23]
```

alonso@LAPTOP-JKJ2KGSC:/home\$

alonso@LAPTOP-JKJ2KGSC:/home\$

alonso@LAPTOP-JKJ2KGSC:/home\$ gpg --full-gen-key

gpg (GnuPG) 2.4.4; Copyright (C) 2024 g10 Code GmbH
This is free software: you are free to change and redistribute it.
There is NO WARRANTY, to the extent permitted by law.

Please select what kind of key you want:

- (1) RSA and RSA
- (2) DSA and Elgamal
- (3) DSA (sign only)
- (4) RSA (sign only)
- (9) ECC (sign and encrypt) *default*
- (10) ECC (sign only)
- (14) Existing key from card

Your selection? 3

DSA keys may be between 1024 and 3072 bits long.

What keysize do you want? (2048) 2048

Requested keysize is 2048 bits

Please specify how long the key should be valid.

0 = key does not expire

<n> = key expires in n days

<n>w = key expires in n weeks

<n>m = key expires in n months

<n>y = key expires in n years

Key is valid for? (0) 1w

Key expires at Sun Nov 23 16:28:31 2025 CST

Is this correct? (y/N) y

GnuPG needs to construct a user ID to identify your key.

Real name: dsa3key

Email address: ismael@gmail.com

Comment: Llave dsa 3 para tarea

You selected this USER-ID:

"dsa3key (Llave dsa 3 para tarea) <ismael@gmail.com>"

Change (N)ame, (C)omment, (E)mail or (O)kay/(Q)uit? o

We need to generate a lot of random bytes. It is a good idea to perform
some other action (type on the keyboard, move the mouse, utilize the
disks) during the prime generation; this gives the random number
generator a better chance to gain enough entropy.

gpg: WARNING: some OpenPGP programs can't handle a DSA key with this digest size

gpg: revocation certificate stored as '/home/alonso/.gnupg/openpgp-revocs.d/FFB17EA257B40134D42AFF28716B53E452270354.rev'

public and secret key created and signed.

public and secret key created and signed.

Note that this key cannot be used for encryption. You may want to use
the command "--edit-key" to generate a subkey for this purpose.

```
pub   dsa2048 2025-11-16 [SC] [expires: 2025-11-23]
      FFB17EA257B40134D42AFF28716B53E452270354
uid    dsa3key (Llave dsa 3 para tarea) <ismael@gmail.com>
```

alonso@LAPTOP-JKJ2KGSC:/home\$

```
alonso@LAPTOP-JKJ2KGSC:/home$ gpg --full-gen-key
gpg (GnuPG) 2.4.4; Copyright (C) 2024 g10 Code GmbH
This is free software: you are free to change and redistribute it.
There is NO WARRANTY, to the extent permitted by law.

Please select what kind of key you want:
(1) RSA and RSA
(2) DSA and Elgamal
(3) DSA (sign only)
(4) RSA (sign only)
(9) ECC (sign and encrypt) *default*
(10) ECC (sign only)
(14) Existing key from card
Your selection? 4
RSA keys may be between 1024 and 4096 bits long.
What keysize do you want? (3072) 3072
Requested keysize is 3072 bits
Please specify how long the key should be valid.
    0 = key does not expire
    <n> = key expires in n days
    <n>w = key expires in n weeks
    <n>m = key expires in n months
    <n>y = key expires in n years
Key is valid for? (0) 1w
Key expires at Sun Nov 23 16:32:58 2025 CST
Is this correct? (y/N) y

GnuPG needs to construct a user ID to identify your key.

Real name: rsa4key
Email address: urrieta@gmail.com
Comment: Llave rsa 4 para tarea
You selected this USER-ID:
    "rsa4key (Llave rsa 4 para tarea) <urrieta@gmail.com>"

Change (N)ame, (C)omment, (E)mail or (O)kay/(Q)uit? o
We need to generate a lot of random bytes. It is a good idea to perform
some other action (type on the keyboard, move the mouse, utilize the
disks) during the prime generation; this gives the random number
generator a better chance to gain enough entropy.
gpg: revocation certificate stored as '/home/alonso/.gnupg/openpgp-revocs.d/E8682BE451FB38DC96AA73101608447B0FEFE779.rev'
public and secret key created and signed.

Note that this key cannot be used for encryption. You may want to use
the command "--edit-key" to generate a subkey for this purpose.
pub   rsa3072 2025-11-16 [SC] [expires: 2025-11-23]
      E8682BE451FB38DC96AA73101608447B0FEFE779
uid    rsa4key (Llave rsa 4 para tarea) <urrieta@gmail.com>

alonso@LAPTOP-JKJ2KGSC:/home$
```

```
alonso@LAPTOP-JKJ2KGSC:/home$ gpg --full-gen-key
gpg (GnuPG) 2.4.4; Copyright (C) 2024 g10 Code GmbH
This is free software: you are free to change and redistribute it.
There is NO WARRANTY, to the extent permitted by law.

Please select what kind of key you want:
(1) RSA and RSA
(2) DSA and Elgamal
(3) DSA (sign only)
(4) RSA (sign only)
(9) ECC (sign and encrypt) *default*
(10) ECC (sign only)
(14) Existing key from card
Your selection? 9
Please select which elliptic curve you want:
(1) Curve 25519 *default*
(4) NIST P-384
(6) Brainpool P-256
Your selection? 1
Please specify how long the key should be valid.
  0 = key does not expire
  <n> = key expires in n days
  <n>w = key expires in n weeks
  <n>m = key expires in n months
  <n>y = key expires in n years
Key is valid for? (0) 1w
Key expires at Sun Nov 23 16:36:26 2025 CST
Is this correct? (y/N) y

GnuPG needs to construct a user ID to identify your key.

Real name: ecc9key
Email address: rangel@gmail.com
Comment: Llave ecc 9 para tarea
You selected this USER-ID:
  "ecc9key (Llave ecc 9 para tarea) <rangel@gmail.com>"

Change (N)ame, (C)omment, (E)mail or (O)kay/(Q)uit? o
We need to generate a lot of random bytes. It is a good idea to perform
some other action (type on the keyboard, move the mouse, utilize the
disks) during the prime generation; this gives the random number
generator a better chance to gain enough entropy.
We need to generate a lot of random bytes. It is a good idea to perform

public and secret key created and signed.

pub  ed25519 2025-11-16 [SC] [expires: 2025-11-23]
     E0DCB1A2C8C4C262EABE58AFE3386B7D995A58A7
uid  ecc9key (Llave ecc 9 para tarea) <rangel@gmail.com>
sub  cv25519 2025-11-16 [E] [expires: 2025-11-23]

alonso@LAPTOP-JKJ2KGSC:/home$
```

```
alonso@LAPTOP-JKJ2KGSC: /home
alonso@LAPTOP-JKJ2KGSC: /home$ gpg --full-gen-key
gpg (GnuPG) 2.4.4; Copyright (C) 2024 g10 Code GmbH
This is free software: you are free to change and redistribute it.
There is NO WARRANTY, to the extent permitted by law.

Please select what kind of key you want:
  (1) RSA and RSA
  (2) DSA and Elgamal
  (3) DSA (sign only)
  (4) RSA (sign only)
  (9) ECC (sign and encrypt) *default*
  (10) ECC (sign only)
  (14) Existing key from card
Your selection? 10
Please select which elliptic curve you want:
  (1) Curve 25519 *default*
  (4) NIST P-384
  (6) Brainpool P-256
Your selection? 1
Please specify how long the key should be valid.
  0 = key does not expire
  <n> = key expires in n days
  <n>w = key expires in n weeks
  <n>m = key expires in n months
  <n>y = key expires in n years
Key is valid for? (0) 1w
Key expires at Sun Nov 23 16:39:11 2025 CST
Is this correct? (y/N) y

GnuPG needs to construct a user ID to identify your key.

Real name: ecc10key
Email address: ecc@gmail.com
Comment: Llave ecc 10 para tarea
You selected this USER-ID:
  "ecc10key (Llave ecc 10 para tarea) <ecc@gmail.com>"

Change (N)ame, (C)omment, (E)mail or (O)kay/(Q)uit? o
We need to generate a lot of random bytes. It is a good idea to perform
some other action (type on the keyboard, move the mouse, utilize the
disks) during the prime generation; this gives the random number
generator a better chance to gain enough entropy.
gpg: revocation certificate stored as '/home/alonso/.gnupg/openpgp-revocs.d/74F20AD324B250B9A8931D0BFE3CFC2C1EF989A5.rev'

public and secret key created and signed.

pub  ed25519 2025-11-16 [SC] [expires: 2025-11-23]
uid  74F20AD324B250B9A8931D0BFE3CFC2C1EF989A5
      ecc10key (Llave ecc 10 para tarea) <ecc@gmail.com>

alonso@LAPTOP-JKJ2KGSC: /home$
```

2. Cuando nos pregunta si los datos son correctos y digitamos "o" que se refiere a "okey" nos dirá que ingresemos una frase/contraseña ara nuestra llave.

```
Please enter the passphrase to
protect your new key

Passphrase: _____

<OK>                                <Cancel>
```

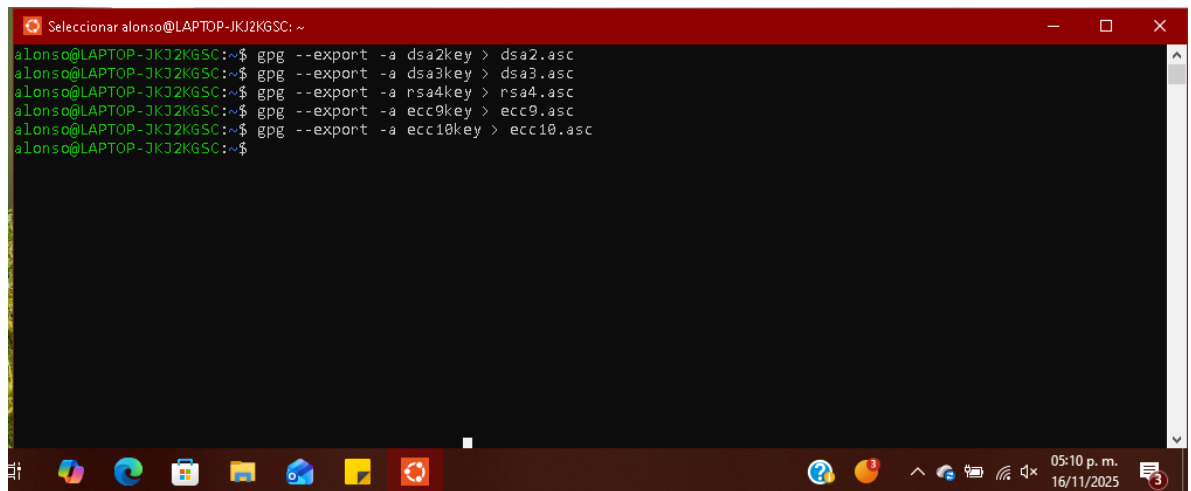
3. Vamos a continuar exportando nuestras llaves con los comandos:
gpg --export -a dsa2key > dsa2.asc

```
gpg --export -a dsa3key > dsa3.asc  
gpg --export -a rsa4key > rsa4.asc  
gpg --export -a ecc9key > ecc9.asc  
gpg --export -a ecc10key > ecc10.asc
```

Donde "dsa2key" es el nombre que le asignamos al usuario de la llave y "dsa2.asc" es el nombre que tendrá nuestro archivo exportado, el comando cambia según la llave.

Además, podemos verificar que nuestros archivos se crearon digitando el comando:

ls -l



```
Seleccionar alonso@LAPTOP-JKJ2KGSC: ~  
alonso@LAPTOP-JKJ2KGSC:~$ gpg --export -a dsa2key > dsa2.asc  
alonso@LAPTOP-JKJ2KGSC:~$ gpg --export -a dsa3key > dsa3.asc  
alonso@LAPTOP-JKJ2KGSC:~$ gpg --export -a rsa4key > rsa4.asc  
alonso@LAPTOP-JKJ2KGSC:~$ gpg --export -a ecc9key > ecc9.asc  
alonso@LAPTOP-JKJ2KGSC:~$ gpg --export -a ecc10key > ecc10.asc  
alonso@LAPTOP-JKJ2KGSC:~$  
alonso@LAPTOP-JKJ2KGSC:~$ ls -l
```

The screenshot shows a terminal window with a red title bar. The terminal displays the execution of five gpg export commands, each creating a .asc file. The prompt then changes to alonso@LAPTOP-JKJ2KGSC:~\$ and the command ls -l is entered. The terminal window has a scrollbar on the right side. The Windows taskbar is visible at the bottom with various icons and a system clock showing 05:10 p. m. on 16/11/2025.

```
alonso@LAPTOP-JKJ2KGSC: ~  
alonso@LAPTOP-JKJ2KGSC:~$ ls -l  
total 52  
-rwxr-xr-x 1 root root 1840 Nov 4 21:25 33.sh  
-rwxr-xr-x 1 alonso alonso 2468 Nov 12 18:41 Eric.asc  
-rwxr-xr-x 1 root root 86 Oct 16 20:25 Hola.sh  
drwxr-xr-x 1 alonso alonso 4096 Oct 23 18:53 SistemasOperativos25B  
-rwxr-xr-x 1 alonso alonso 785 Nov 4 19:47 alerta.log  
-rwxr-xr-x 1 alonso alonso 2525 Nov 12 18:40 alonsokey.asc  
drwxr-xr-x 1 alonso alonso 4096 Oct 18 12:59 backups  
-rwxr-xr-x 1 root root 119 Oct 16 20:29 bucles.sh  
-rwxr-xr-x 1 root root 31 Nov 12 19:20 cancion.txt  
-rwxr-xr-x 1 alonso alonso 112 Nov 12 18:41 cancion.txt.gpg  
-rwxr-xr-x 1 alonso alonso 511 Nov 12 19:23 canciones  
drwxr-xr-x 1 alonso alonso 4096 Oct 21 20:14 carpetas  
-rwxr-xr-x 1 root root 1756 Oct 21 20:06 carpetas.sh  
-rwxr-xr-x 1 root root 225 Oct 16 20:35 case.sh  
-rwxr-xr-x 1 root root 1412 Oct 23 19:02 cesar.py  
-rwxr-xr-x 1 alonso alonso 63 Nov 4 22:42 citas.txt  
-rwxr-xr-x 1 alonso alonso 2274 Oct 12 17:54 comandos91025.txt  
-rwxr-xr-x 1 root root 160 Oct 16 20:22 condi.sh  
drwxr-xr-x 1 alonso alonso 4096 Nov 4 20:21 copia_seguridad  
-rwxr-xr-x 1 root root 109 Oct 16 20:11 dias.sh  
-rwxr-xr-x 1 root root 73 Oct 16 20:27 dir.sh  
-rwxr-xr-x 1 alonso alonso 2359 Nov 16 17:08 dsa2.asc  
-rwxr-xr-x 1 alonso alonso 1469 Nov 16 17:08 dsa3.asc  
-rwxr-xr-x 1 alonso alonso 433 Nov 16 17:09 ecc10.asc  
-rwxr-xr-x 1 alonso alonso 685 Nov 16 17:09 ecc9.asc  
-rwxr-xr-x 1 root root 75 Oct 16 20:30 for.sh  
-rwxr-xr-x 1 root root 72 Oct 16 20:23 fun.sh  
-rwxr-xr-x 1 root root 352 Oct 16 20:08 function.sh  
-rwxr-xr-x 1 root root 263 Oct 16 20:32 ifelse.sh  
drwxr-xr-x 1 root root 4096 Oct 17 20:07 juegos  
drwxr-xr-x 1 root root 4096 Oct 17 20:05 materias  
drwxr-xr-x 1 alonso alonso 4096 Oct 16 20:28 myDir  
drwxr-xr-x 1 root root 4096 Oct 17 20:03 prueba  
-rwxr-xr-x 1 root root 1019 Oct 18 12:59 respaldo.sh  
-rwxr-xr-x 1 alonso alonso 1339 Nov 16 17:08 rsa4.asc  
drwxr-xr-x 1 alonso alonso 4096 Nov 4 22:40 shell  
-rwxr-xr-x 1 root root 177 Oct 16 20:26 suma.sh  
-rwxr-xr-x 1 alonso alonso 29 Nov 4 22:45 ultimo-crontab.txt  
-rwxr-xr-x 1 root root 60 Oct 16 20:14 until.sh  
-rwxr-xr-x 1 root root 60 Oct 16 20:13 while.sh  
alonso@LAPTOP-JKJ2KGSC:~$
```

4. Nuestro compañero de equipo hizo exactamente los mismos pasos anteriores, nos debe pasar sus archivos .asc y su .gpg, para que nosotros importemos sus llaves a nuestro llavero con los comandos:

```
gpg --import EricSoto2_opcion2.asc  
gpg --import EricSoto3_opcion3.asc  
gpg --import EricSoto4_opcion4.asc  
gpg --import EricSoto9_opcion9.asc  
gpg --import EricSoto10_opcion10.asc  
gpg --import LlaveEricSotoL.asc
```



```
alonso@LAPTOP-JKJ2KGSC:~$ gpg --import EricSoto2_opcion2.asc
gpg: key FC3D6C37810CA017: public key "EricSoto2 (llave con opción 2) <ericsoto2@gmail.com>" imported
gpg: Total number processed: 1
gpg:      imported: 1
alonso@LAPTOP-JKJ2KGSC:~$ gpg --import EricSoto3_opcion3.asc
gpg: key 19833A9FA6E441D8: public key "EricSoto3 (llave version 3) <ericsoto3@gmail.com>" imported
gpg: Total number processed: 1
gpg:      imported: 1
alonso@LAPTOP-JKJ2KGSC:~$ gpg --import EricSoto4_opcion4.asc
gpg: key 3072CDAE24A76C36: public key "EricSoto4 (llave version 4) <ericsoto4@gmail.com>" imported
gpg: Total number processed: 1
gpg:      imported: 1
alonso@LAPTOP-JKJ2KGSC:~$ gpg --import EricSoto9_opcion9.asc
gpg: key 5D5590B3F38FF978: public key "EricSoto9 (llave version 9) <ericosoto9@gmail.com>" imported
gpg: Total number processed: 1
gpg:      imported: 1
alonso@LAPTOP-JKJ2KGSC:~$ gpg --import EricSoto10_opcion10.asc
gpg: can't open 'EricSoto10_opcion10.asc': No such file or directory
gpg: Total number processed: 0
alonso@LAPTOP-JKJ2KGSC:~$ gpg --import EricSoto10_opcion10.asc
gpg: key 6330D3C75C0A8CF5: public key "EricSoto10 (llave version 10) <ericsoto10@gmail.com>" imported
gpg: Total number processed: 1
gpg:      imported: 1
alonso@LAPTOP-JKJ2KGSC:~$ gpg --import llaveEricSoto.asc
gpg: key 088080B9E3F96A52: public key "EricSotoL (trabajo 13-11-2025) <erics@gmail.com>" imported
gpg: Total number processed: 1
gpg:      imported: 1
alonso@LAPTOP-JKJ2KGSC:~$
```

5. Podemos comprobar que las llaves de nuestro compañero ahora están en nuestro llavero con el comando:

gpg -k

```
alonso@LAPTOP-JKJ2KGSC: ~
uid      [ unknown] EricSoto2 (llave con opción 2) <ericsoto2@gmail.com>
sub      elg2048 2025-11-17 [E] [expires: 2025-12-17]

pub      dsa2048 2025-11-17 [SC] [expires: 2025-12-17]
        6AC648DDE2115A7B21CAF94119833A9FA6E441D8
uid      [ unknown] EricSoto3 (llave version 3) <ericsoto3@gmail.com>

pub      rsa3072 2025-11-17 [SC] [expires: 2025-12-17]
        D58C103EE7BC00665C3084833072CDAE24A76C36
uid      [ unknown] EricSoto4 (llave version 4) <ericsoto4@gmail.com>

pub      ed25519 2025-11-17 [SC] [expires: 2025-12-17]
        D044AD2ACE25A9875D6851365D5590B3F38FF978
uid      [ unknown] EricSoto9 (llave version 9) <ericosoto9@gmail.com>
sub      cv25519 2025-11-17 [E] [expires: 2025-12-17]

pub      ed25519 2025-11-17 [SC] [expires: 2025-12-17]
        F5A99FEFAA578567D08BA95F6330D3C75C0A8CF5
uid      [ unknown] EricSoto10 (llave version 10) <ericsoto10@gmail.com>

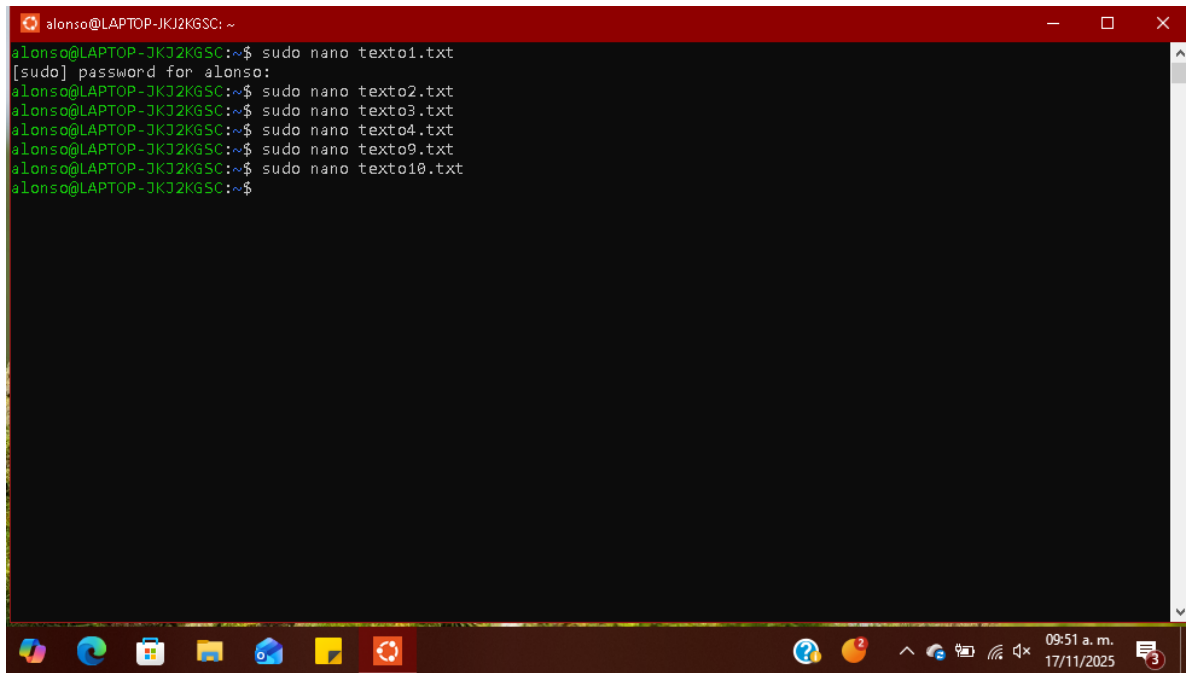
pub      rsa3072 2025-11-13 [SC] [expires: 2025-11-20]
        C63B3C975C39B3C58BD53436088089B9E3F96A52
uid      [ unknown] EricSotoL (trabajo 13-11-2025) <erics@gmail.com>
sub      rsa3072 2025-11-13 [E] [expires: 2025-11-20]

alonso@LAPTOP-JKJ2KGSC:~$
```

6. Anteriormente mencione unos archivo .gpg, este se refiere a un archivo de texto que vamos a encriptar, usaremos los comandos siguientes:

sudo nano texto1.txt

```
sudo nano texto2.txt
sudo nano texto3.txt
sudo nano texto4.txt
sudo nano texto9.txt
sudo nano texto10.txt
```



```
alonso@LAPTOP-JKJ2KGSC: ~
alonso@LAPTOP-JKJ2KGSC:~$ sudo nano texto1.txt
[sudo] password for alonso:
alonso@LAPTOP-JKJ2KGSC:~$ sudo nano texto2.txt
alonso@LAPTOP-JKJ2KGSC:~$ sudo nano texto3.txt
alonso@LAPTOP-JKJ2KGSC:~$ sudo nano texto4.txt
alonso@LAPTOP-JKJ2KGSC:~$ sudo nano texto9.txt
alonso@LAPTOP-JKJ2KGSC:~$ sudo nano texto10.txt
alonso@LAPTOP-JKJ2KGSC:~$
```

7. Continuando con el paso anterior, dichos archivos serán encriptados con ayuda del comando:

```
gpg -u <TU_LLAVE> -se -r <DESTINATARIO_LLAVE> <ARCHIVO.txt>
```

Este comando lo que hará será:

- -u <TU_LLAVE>: Especifica tu clave para firmar
- -s: Firma el archivo
- -e: Cifra el archivo
- -r <DESTINATARIO_LLAVE>: Especifica la clave del destinatario para cifrado
- <ARCHIVO.txt>: El archivo a procesar

```
slonso@LAPTOP-IKJ2K6SC:~$
slonso@LAPTOP-IKJ2K6SC:~$ gpg -u rsa1key -se -r C63B3C975C39B3C58B053436080809B9E3F96A52 texto1.txt
gpg: 4652FA08735B7B65: There is no assurance this key belongs to the named user

sub rsa3072/4652FA08735B7B65 2025-11-13 EricSotoL (trabajo 13-11-2025) <erics@gmail.com>
Primary key fingerprint: C63B 3C97 5C39 B3C5 8B05 3436 0808 09B9 E3F9 6A52
Subkey fingerprint: 75F6 A93A E3FB 7A28 D610 B20A 4652 FA08 735B 7B65

It is NOT certain that the key belongs to the person named
in the user ID. If you *really* know what you are doing,
you may answer the next question with yes.

Use this key anyway? (y/N) y
slonso@LAPTOP-IKJ2K6SC:~$ gpg -u ds2key -se -r 0B387E6241370FE3602C1120FC3D6C37818CA017 texto2.txt
gpg: 535C7250E4D1F663: There is no assurance this key belongs to the named user

sub elg2048/535C7250E4D1F663 2025-11-17 EricSoto2 (llave con opc1dn 2) <ericsoto2@gmail.com>
Primary key fingerprint: 9B38 7E62 4137 0FE3 602C 1120 FC3D 6C37 810C A017
Subkey fingerprint: A6E6 274F 8B05 E24E 7ECA 5ACC 935C 7250 E4D1 F663

It is NOT certain that the key belongs to the person named
in the user ID. If you *really* know what you are doing,
you may answer the next question with yes.

Use this key anyway? (y/N) y
slonso@LAPTOP-IKJ2K6SC:~$ gpg -u ds3key -se -r 6AC6480DE2115A7B21CAF94119833A9FA6E4410B texto3.txt
gpg: 6AC6480DE2115A7B21CAF94119833A9FA6E4410B: skipped: Unusable public key
gpg: texto3.txt: sign+encrypt failed: Unusable public key
slonso@LAPTOP-IKJ2K6SC:~$ gpg -u ds3key -se -r C63B3C975C39B3C58B053436080809B9E3F96A52 texto3.txt
gpg: 4652FA08735B7B65: There is no assurance this key belongs to the named user

sub rsa3072/4652FA08735B7B65 2025-11-13 EricSotoL (trabajo 13-11-2025) <erics@gmail.com>
Primary key fingerprint: C63B 3C97 5C39 B3C5 8B05 3436 0808 09B9 E3F9 6A52
Subkey fingerprint: 75F6 A93A E3FB 7A28 D610 B20A 4652 FA08 735B 7B65

It is NOT certain that the key belongs to the person named
in the user ID. If you *really* know what you are doing,
you may answer the next question with yes.

Use this key anyway? (y/N) y
slonso@LAPTOP-IKJ2K6SC:~$ gpg -u rs4key -se -r D58C183EE7BC08665C3884833872CDAE24A76C30 texto4.txt
gpg: D58C183EE7BC08665C3884833872CDAE24A76C30: skipped: Unusable public key
gpg: texto4.txt: sign+encrypt failed: Unusable public key
slonso@LAPTOP-IKJ2K6SC:~$ gpg -u rs4key -se -r C63B3C975C39B3C58B053436080809B9E3F96A52 texto4.txt
gpg: 4652FA08735B7B65: There is no assurance this key belongs to the named user
```

```
slonso@LAPTOP-IKJ2K6SC:~$
slonso@LAPTOP-IKJ2K6SC:~$ gpg -u rs4key -se -r C63B3C975C39B3C58B053436080809B9E3F96A52 texto4.txt
gpg: 4652FA08735B7B65: There is no assurance this key belongs to the named user

sub rsa3072/4652FA08735B7B65 2025-11-13 EricSotoL (trabajo 13-11-2025) <erics@gmail.com>
Primary key fingerprint: C63B 3C97 5C39 B3C5 8B05 3436 0808 09B9 E3F9 6A52
Subkey fingerprint: 75F6 A93A E3FB 7A28 D610 B20A 4652 FA08 735B 7B65

It is NOT certain that the key belongs to the person named
in the user ID. If you *really* know what you are doing,
you may answer the next question with yes.

Use this key anyway? (y/N) y
slonso@LAPTOP-IKJ2K6SC:~$ gpg -u ecc9key -se -r D044AD2ACE25A9875D60851365D5590B3F38FF970 texto5.txt
gpg: BCE308F9FCBA2DE4: There is no assurance this key belongs to the named user

sub cv25519/BCE308F9FCBA2DE4 2025-11-17 EricSoto9 (llave version 9) <ericosoto9@gmail.com>
Primary key fingerprint: D044 AD2A CE25 A987 5D68 5136 5D55 90B3 F38F F978
Subkey fingerprint: 9F7A B513 B058 C71E F28A BB41 BCE3 08F9 FCBA 2DE4

It is NOT certain that the key belongs to the person named
in the user ID. If you *really* know what you are doing,
you may answer the next question with yes.

Use this key anyway? (y/N) y
gpg: can't open 'texto5.txt': No such file or directory
gpg: texto5.txt: sign+encrypt failed: No such file or directory
slonso@LAPTOP-IKJ2K6SC:~$ gpg -u ecc10key -se -r F5A99FEFAA578567D08BA95F633803C75C0A8CF5 texto6.txt
gpg: F5A99FEFAA578567D08BA95F633803C75C0A8CF5: skipped: Unusable public key
gpg: texto6.txt: sign+encrypt failed: Unusable public key
slonso@LAPTOP-IKJ2K6SC:~$ gpg -u ecc10key -se -r C63B3C975C39B3C58B053436080809B9E3F96A52 texto6.txt
gpg: 4652FA08735B7B65: There is no assurance this key belongs to the named user

sub rsa3072/4652FA08735B7B65 2025-11-13 EricSotoL (trabajo 13-11-2025) <erics@gmail.com>
Primary key fingerprint: C63B 3C97 5C39 B3C5 8B05 3436 0808 09B9 E3F9 6A52
Subkey fingerprint: 75F6 A93A E3FB 7A28 D610 B20A 4652 FA08 735B 7B65

It is NOT certain that the key belongs to the person named
in the user ID. If you *really* know what you are doing,
you may answer the next question with yes.

Use this key anyway? (y/N) y
gpg: can't open 'texto6.txt': No such file or directory
gpg: texto6.txt: sign+encrypt failed: No such file or directory
slonso@LAPTOP-IKJ2K6SC:~$
```

```
alonso@LAPTOP-JKJ2KGSC: ~  
alonso@LAPTOP-JKJ2KGSC:~$ gpg -u ecc9key -se -r D044AD2ACE25A9875D6851365D5590B3F38FF978 texto9.txt  
gpg: BCE308F9FCBA2DE4: There is no assurance this key belongs to the named user  
  
sub cv25519/BCE308F9FCBA2DE4 2025-11-17 EricSoto9 (llave version 9) <ericosoto9@gmail.com>  
Primary key fingerprint: D044 AD2A CE25 A987 5D68 5136 5D55 00B3 F38F F978  
Subkey fingerprint: 9F7A B513 B058 C71E F28A BB41 BCE3 08F9 FCBA 2DE4  
  
It is NOT certain that the key belongs to the person named  
in the user ID. If you *really* know what you are doing,  
you may answer the next question with yes.  
  
Use this key anyway? (y/N) y  
alonso@LAPTOP-JKJ2KGSC:~$
```

```
alonso@LAPTOP-JKJ2KGSC: ~  
alonso@LAPTOP-JKJ2KGSC:~$ gpg -u ecc10key -se -r C63B3C975C39B3C58BD53436088089B9E3F96A52 texto10.txt  
gpg: 4652FA08735B7B65: There is no assurance this key belongs to the named user  
  
sub rsa3072/4652FA08735B7B65 2025-11-13 EricSotoL (trabajo 13-11-2025) <erics@gmail.com>  
Primary key fingerprint: C63B 3C97 5C39 B3C5 8BD5 3436 0880 89B9 E3F9 6A52  
Subkey fingerprint: 75F6 A93A E3FB 7A28 D610 B20A 4652 FA08 735B 7B65  
  
It is NOT certain that the key belongs to the person named  
in the user ID. If you *really* know what you are doing,  
you may answer the next question with yes.  
  
Use this key anyway? (y/N) y  
alonso@LAPTOP-JKJ2KGSC:~$
```

8. Aplicamos un `ls -l` y verificamos que nuestros archivos de texto fueron encriptados correctamente

```
alonso@LAPTOP-JKJ2KGSC: ~  
-rw-r--r-- 1 alonso alonso 790 Nov 17 08:59 mensaje3.txt.gpg  
-rw-r--r-- 1 alonso alonso 1126 Nov 17 08:59 mensaje4.txt.gpg  
-rw-r--r-- 1 alonso alonso 356 Nov 17 08:59 mensaje9.txt.gpg  
drwxr-xr-x 1 alonso alonso 4096 Oct 16 20:28 myDir  
drwxr-xr-x 1 root root 4096 Oct 17 20:03 prueba  
-rw-r--r-- 1 root root 1019 Oct 18 12:59 respaldo.sh  
-rw-r--r-- 1 alonso alonso 2480 Nov 17 09:32 rsa1.asc  
-rw-r--r-- 1 alonso alonso 1339 Nov 16 17:08 rsa4.asc  
drwxr-xr-x 1 alonso alonso 4096 Nov 4 22:40 shell  
-rw-r--r-- 1 root root 177 Oct 16 20:26 suma.sh  
-rw-r--r-- 1 root root 23 Nov 17 09:49 texto1.txt  
-rw-r--r-- 1 alonso alonso 975 Nov 17 10:13 texto1.txt.gpg  
-rw-r--r-- 1 root root 38 Nov 17 09:51 texto10.txt  
-rw-r--r-- 1 alonso alonso 660 Nov 17 10:24 texto10.txt.gpg  
-rw-r--r-- 1 root root 11 Nov 17 09:49 texto2.txt  
-rw-r--r-- 1 alonso alonso 765 Nov 17 10:14 texto2.txt.gpg  
-rw-r--r-- 1 root root 33 Nov 17 09:50 texto3.txt  
-rw-r--r-- 1 alonso alonso 653 Nov 17 10:16 texto3.txt.gpg  
-rw-r--r-- 1 root root 18 Nov 17 09:50 texto4.txt  
-rw-r--r-- 1 alonso alonso 974 Nov 17 10:17 texto4.txt.gpg  
-rw-r--r-- 1 root root 25 Nov 17 09:51 texto9.txt  
-rw-r--r-- 1 alonso alonso 346 Nov 17 10:28 texto9.txt.gpg  
-rw-r--r-- 1 alonso alonso 29 Nov 4 22:45 ultimo-crontab.txt  
-rw-r--r-- 1 root root 60 Oct 16 20:14 until.sh  
-rw-r--r-- 1 root root 60 Oct 16 20:13 while.sh  
alonso@LAPTOP-JKJ2KGSC:~$
```

9. Para finalizar vamos a descifrar los mensajes de nuestro compañero, para realizar esto vamos a ocupar el comando:

`gpg --decrypt mensajeX.txt.gpg > mensajeX.txt`

```
alonso@LAPTOP-JKJ2KGSC:~$ gpg --decrypt mensaje1.txt.gpg > mensaje1.txt  
gpg: encrypted with elg2048 key, ID 381025C2A99FB5CC, created 2025-11-16  
"dsa2key (llave DSA 2 para reporte) <alonso@gmail.com>"  
gpg: Signature made Mon Nov 17 03:16:10 2025 CST  
gpg: using RSA key C63B3C975C39B3C58B053436080809B9E3F96A52  
gpg: issuer "ericsoto@gmail.com"  
gpg: Good signature from "EricSoto2 (trabajo 13-11-2025) <erics@gmail.com>" [unknown]  
gpg: WARNING: The key's User ID is not certified with a trusted signature!  
gpg: There is no indication that the signature belongs to the owner.  
Primary key fingerprint: C63B 3C97 5C39 B3C5 8B05 3436 0808 09B9 E3F9 6A52  
alonso@LAPTOP-JKJ2KGSC:~$ gpg --decrypt mensaje2.txt.gpg > mensaje2.txt  
gpg: encrypted with elg2048 key, ID 381025C2A99FB5CC, created 2025-11-16  
"dsa2key (llave DSA 2 para reporte) <alonso@gmail.com>"  
gpg: Signature made Mon Nov 17 03:19:50 2025 CST  
gpg: using DSA key 0B387E6241370FE3602C1128FC3D6C37818CA017  
gpg: issuer "ericsoto2@gmail.com"  
gpg: Good signature from "EricSoto2 (llave con opción 2) <ericsoto2@gmail.com>" [unknown]  
gpg: WARNING: The key's User ID is not certified with a trusted signature!  
gpg: There is no indication that the signature belongs to the owner.  
Primary key fingerprint: 0B38 7E62 4137 0FE3 602C 1128 FC3D 6C37 810C A017  
alonso@LAPTOP-JKJ2KGSC:~$ gpg --decrypt mensaje3.txt.gpg > mensaje3.txt  
gpg: encrypted with elg2048 key, ID 381025C2A99FB5CC, created 2025-11-16  
"dsa2key (llave DSA 2 para reporte) <alonso@gmail.com>"  
gpg: Signature made Mon Nov 17 03:29:04 2025 CST  
gpg: using DSA key 6AC648DDE2115A7B21CAF94119833A9FA6E4410B  
gpg: issuer "ericsoto3@gmail.com"  
gpg: Good signature from "EricSoto3 (llave version 3) <ericsoto3@gmail.com>" [unknown]  
gpg: WARNING: The key's User ID is not certified with a trusted signature!  
gpg: There is no indication that the signature belongs to the owner.  
Primary key fingerprint: 6AC6 48DD E211 5A7B 21CA F941 1983 3A9F A6E4 410B  
alonso@LAPTOP-JKJ2KGSC:~$ gpg --decrypt mensaje4.txt.gpg > mensaje4.txt  
gpg: encrypted with elg2048 key, ID 381025C2A99FB5CC, created 2025-11-16  
"dsa2key (llave DSA 2 para reporte) <alonso@gmail.com>"  
gpg: Signature made Mon Nov 17 03:29:55 2025 CST  
gpg: using RSA key D58C103EE7B000665C3884833072CDAE24A76C36  
gpg: issuer "ericsoto4@gmail.com"  
gpg: Good signature from "EricSoto4 (llave version 4) <ericsoto4@gmail.com>" [unknown]  
gpg: WARNING: The key's User ID is not certified with a trusted signature!  
gpg: There is no indication that the signature belongs to the owner.  
Primary key fingerprint: D58C 103E E7BC 0066 5C30 8483 3072 CDAE 24A7 6C36  
alonso@LAPTOP-JKJ2KGSC:~$ gpg --decrypt mensaje9.txt.gpg > mensaje9.txt  
gpg: encrypted with cv25519 key, ID 68A0CFBDB8562024, created 2025-11-16  
"ecc9key (llave ecc 9 para tarea) <anagel@gmail.com>"  
gpg: Signature made Mon Nov 17 03:30:51 2025 CST
```

```
alonso@LAPTOP-JKJ2KGSC:~$ gpg --decrypt mensaje9.txt.gpg > mensaje9.txt
gpg: encrypted with cv25519 key, ID 68A0CFB0B8562D24, created 2025-11-16
      "ecc9key (llave ecc 9 para tarea) <rangel@gmail.com>"
gpg: Signature made Mon Nov 17 03:30:51 2025 CST
      using ED05A key D044AD2ACE25A9875D6851365D5590B3F38FF978
      issuer "ericosoto9@gmail.com"
gpg: Good signature from "EricSoto9 (llave version 9) <ericosoto9@gmail.com>" [unknown]
gpg: WARNING: The key's User ID is not certified with a trusted signature!
gpg:       There is no indication that the signature belongs to the owner.
Primary key fingerprint: D044 AD2A CE25 A987 5D68 5136 5D55 90B3 F38F F978
alonso@LAPTOP-JKJ2KGSC:~$ gpg --decrypt mensaje9.txt.gpg > mensaje9.txt
gpg: encrypted with cv25519 key, ID 68A0CFB0B8562D24, created 2025-11-16
      "ecc9key (llave ecc 9 para tarea) <rangel@gmail.com>"
gpg: Signature made Mon Nov 17 03:30:51 2025 CST
      using ED05A key D044AD2ACE25A9875D6851365D5590B3F38FF978
      issuer "ericosoto9@gmail.com"
gpg: Good signature from "EricSoto9 (llave version 9) <ericosoto9@gmail.com>" [unknown]
gpg: WARNING: The key's User ID is not certified with a trusted signature!
gpg:       There is no indication that the signature belongs to the owner.
Primary key fingerprint: D044 AD2A CE25 A987 5D68 5136 5D55 90B3 F38F F978
alonso@LAPTOP-JKJ2KGSC:~$ gpg --decrypt mensaje10.txt.gpg > mensaje10.txt
gpg: encrypted with elg2048 key, ID 381025C2A99FB5CC, created 2025-11-16
      "dsa2key (llave DSA 2 para reporte) <alonso@gmail.com>"
gpg: Signature made Mon Nov 17 03:31:52 2025 CST
      using ED05A key F5A99FEFAA578567D0BB8A95F6330D3C75C0A8CF5
      issuer "ericosoto10@gmail.com"
gpg: Good signature from "EricSoto10 (llave version 10) <ericosoto10@gmail.com>" [unknown]
gpg: WARNING: The key's User ID is not certified with a trusted signature!
gpg:       There is no indication that the signature belongs to the owner.
Primary key fingerprint: F5A9 9FEF AA57 8567 D0BB A95F 6330 D3C7 5C0A 8CF5
alonso@LAPTOP-JKJ2KGSC:~$
```

10. Ahora si podemos leer el contenido de cada documento txt.

```
alonso@LAPTOP-JKJ2KGSC: ~
alonso@LAPTOP-JKJ2KGSC:~$ cat mensaje1.txt mensaje2.txt mensaje3.txt mensaje4.txt mensaje9.txt mensaje10.txt
Mensaje para la llave tipo 1 (keyAlonso)
Mensaje para la llave tipo 2 (dsa2)
Mensaje para la llave tipo 3 (dsa3)
Mensaje para la llave tipo 4 (rsa4)
Mensaje para la llave tipo 9 (ecc9)
Mensaje para la llave tipo 10 (ecc10)
alonso@LAPTOP-JKJ2KGSC:~$
```